7.2 (1) √ (2) × (3) √ (4) ×

7.6. (1) 由起振条件 $R_f + R_w' \geq 2R$.

$\therefore R_w' \geq 2k\Omega$.

(2). $(f_0)_{max} = \frac{1}{2\pi R_1 C} \approx 1.6kHz$

$(f_0)_{min} = \frac{1}{2\pi C(R_1+R_2)} \approx 145Hz$.

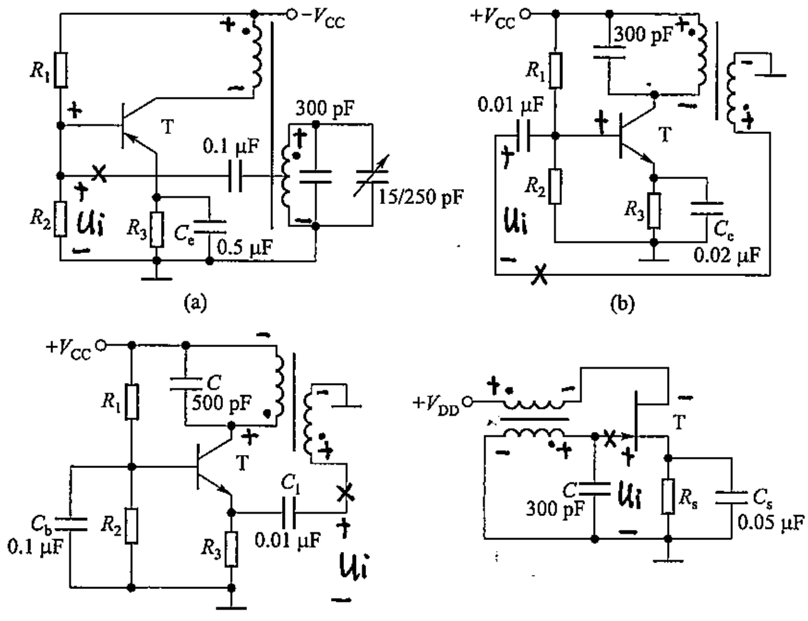
7.7 (1). $\because R_f = 2R_1$.

$\therefore U_o = U_f + U_1 = \frac{3}{2}U_Z = \pm 9V$

有效值为 $U_o = \frac{9}{\sqrt{2}}V \approx 6.36V$.

(2). $f = \frac{1}{2\pi CR} \approx 9.95Hz$.

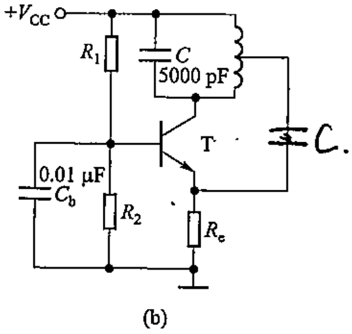
7.9



(a) (b) (c) (d)

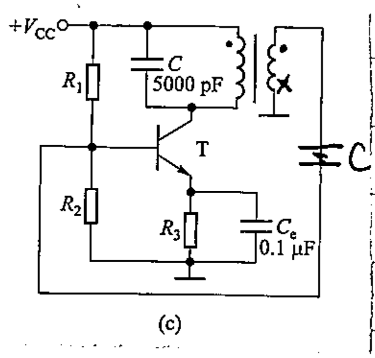
7.10.(a)可能 (b)不可能 (c)不可能 (d)可能.

7.11 (b)



(b)

(c)



(c)

7.12(a) 选频网络：C和L.

正反馈网络：R_3，C_2和R_W.

负反馈网络：C和L.

满足振荡条件

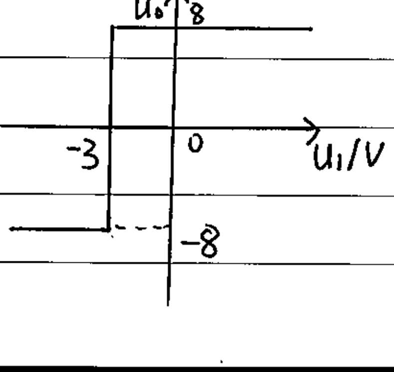
(b) ~~频~~选频网络：C_2和L.

正反馈~~电路~~网络：R_4，C_2和L.

负反馈~~电~~网络：R8.

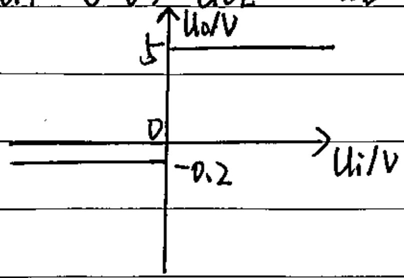
满足振荡条件.

7.13 (a) $u_o = \pm U_Z = \pm 8V$. $U_T = -\frac{R_2}{R_1}\cdot U_{REF} = -3V$.



(b) $U_T = 0\,V$, $U_{OL} = -U_D = -0.2\,V$, $U_{OH} = +U_Z = \pm 5\,V$.

∴

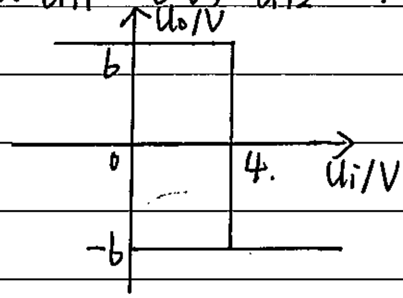


(c). $U_O = \pm U_T = \pm 6\,V$.

$$U_P = \frac{R_1}{R_1+R_2}U_O + \frac{R_2}{R_1+R_2}\cdot U_{REF}$$ 式

令 $U_P = U_N = U_I$, 则 $U_N = \frac{1}{3}U_O + 2$.

∴ $U_{T1} = 0\,V$, $U_{T2} = 4\,V$.

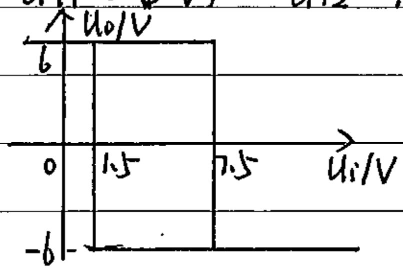


(d). $U_O = \pm U_Z = \pm 6\,V$.

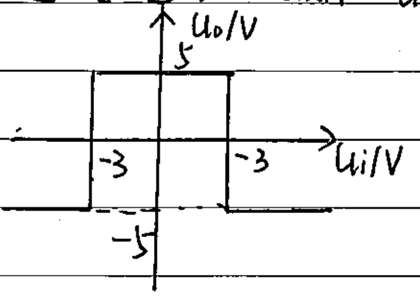
$$U_P = \frac{R_2}{R_1+R_2}U_I + \frac{R_1}{R_1+R_2}U_O = \frac{1}{3}U_O + \frac{2}{3}U_I$$

令 $U_P = U_N = 3V$. $U_I = \cancel{9 = U_O}\ \frac{1}{2}(9 - U_O)$

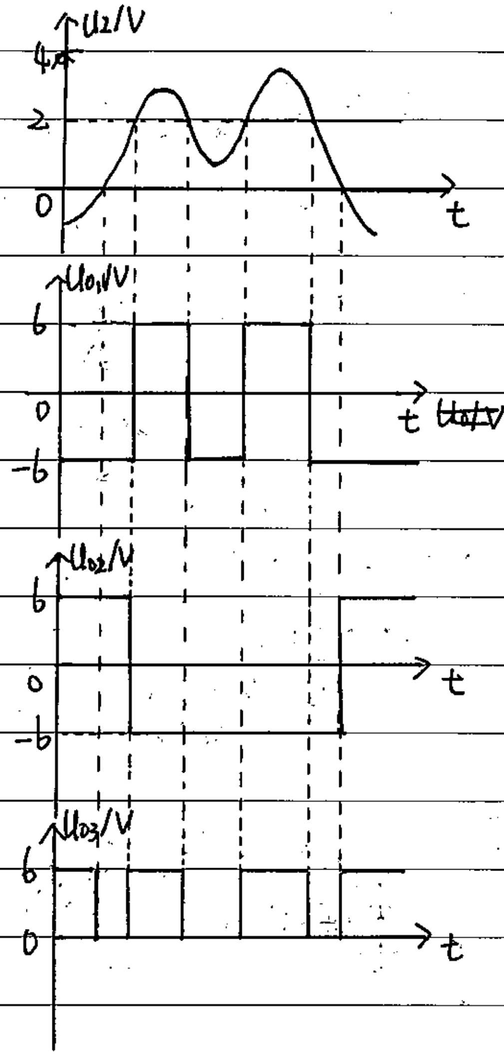
∴ $U_{T1} = \cancel{3}\ 1.5\,V$, $U_{T2} = 7.5\,V$.



(e). $u_o = \pm U_Z = \pm 5V$. $U_{RH} = U_T = 3V$, $U_{RL} = -3V$.

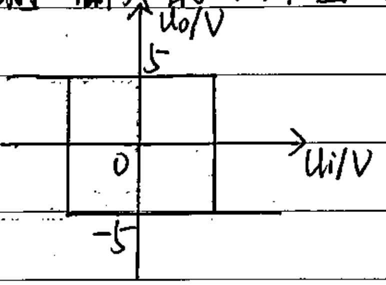


7.14. (a).

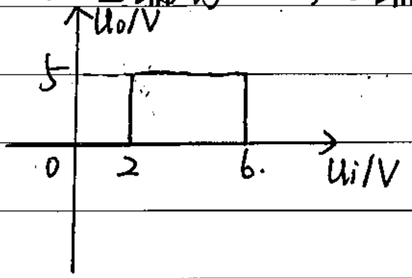


7.15 (b) 2端为"−", 3端为"+",

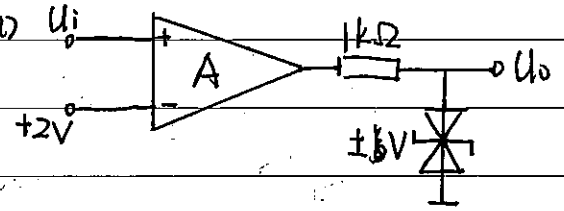
反相输入的滞回比较器.



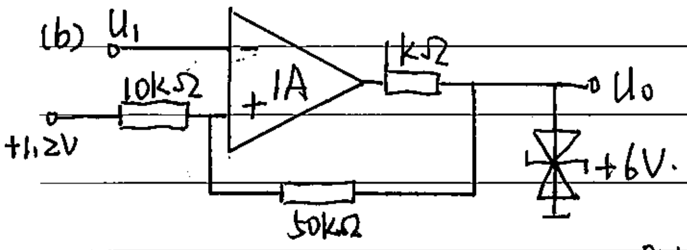
(c). ~~2~~ 2端为"−", 3端为"+".



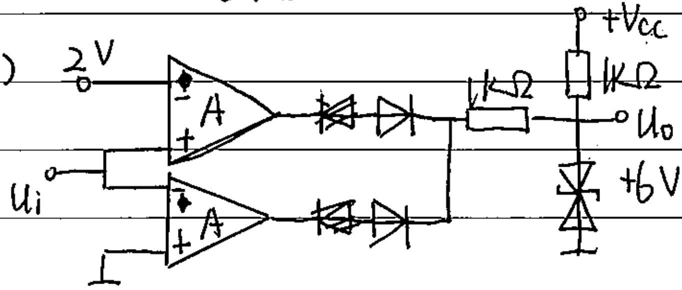
7.17 (a)



(b)



(c)



7.20. ②,①,③；②,①,②；①②②

7.22

(1) $\boxed{温度信号} \rightarrow \boxed{温度传感器} \rightarrow \boxed{放大电路} \rightarrow \boxed{压控振荡电路} \rightarrow \boxed{频率计}$

(2). 交流电压信号 $\rightarrow \boxed{放大电路} \rightarrow \boxed{精密整流电路} \rightarrow \boxed{低通滤波电路} \rightarrow \boxed{压控振荡电路} \rightarrow \boxed{频率计}$

(3) $\boxed{文氏振荡电桥} \rightarrow \boxed{C-ACV转换电路} \rightarrow \boxed{精密整流电路} \rightarrow \boxed{带通滤波器} \rightarrow \boxed{压控振荡电路} \rightarrow \boxed{频率计}$

7.23 (a) 当 $u_I > 0$ 时，D_1, D_4 导通，D_2, D_3 截止，A_2 为电压跟随器，$u_O = u_I$；

当 $u_I < 0$ 时，D_2, D_3 导通，D_1, D_4 ~~导通~~ 截止，A_1 为反相比例运算电路，则 $u_O = -\frac{R_4}{R_1} u_I = -u_I$

$\therefore u_O = |u_I|$

(b). 当 $u_I > 0$ 时. D_1, D_4 导通，D_2, D_3 截止，

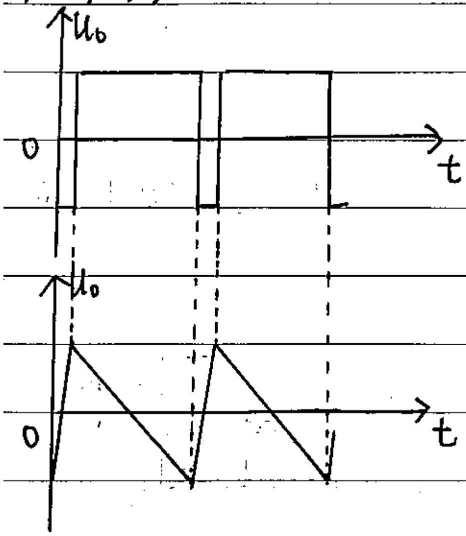
$u_O = \frac{R_L}{R_1} u_I$.

当 $u_I < 0$ 时，D_1, D_4 截止，D_2, D_3 导通.

$u_O = -\frac{R_L}{R_1} u_I$

$\therefore u_O = \frac{R_L}{R_1} |u_I|$.

7.24 (1).



(2) $U_T = \pm U_Z = \pm 8V$.

$\because -U_T = -\frac{1}{R_1 C} u_1 T + U_T$

$\therefore T \approx \frac{2U_T R_1 C}{u_1}$

$\therefore f = \frac{u_1}{2U_T R_1 C} \approx 62.5 u_I$